

N8N • PINECONE • GPT-4.1-MINI

Legal HR RAG Bot

A two-tier RAG pipeline for HR and legal self-service, delivered over Telegram

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Problem & context

Employees regularly ask questions about HR policies and regulations — the answers are scattered across internal documents and legal texts. Manual lookup costs HR and legal staff time, and answer quality depends on who happens to respond.

Speed

Most questions are routine. They need an instant answer, not a wait for manual handling.

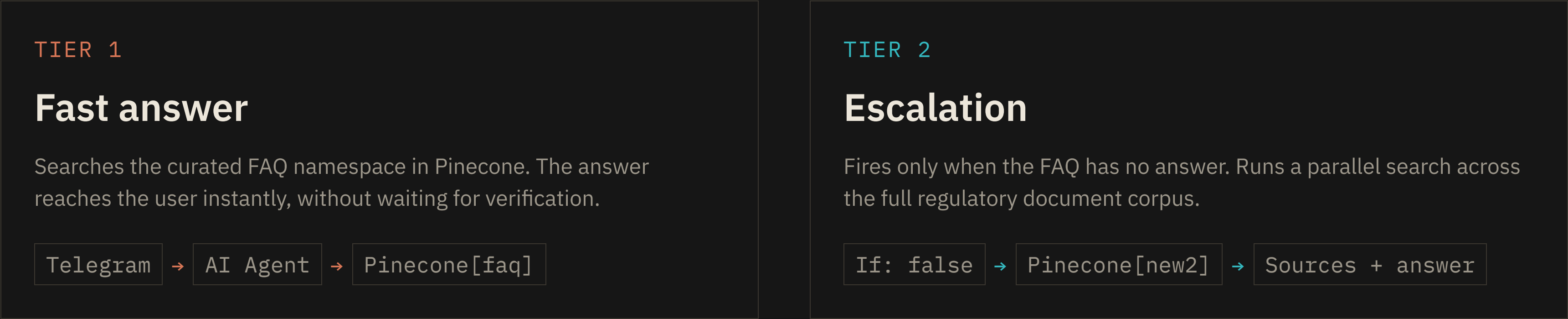
Trustworthiness

Answers must be grounded in real documents, not the model's general knowledge.

Coverage

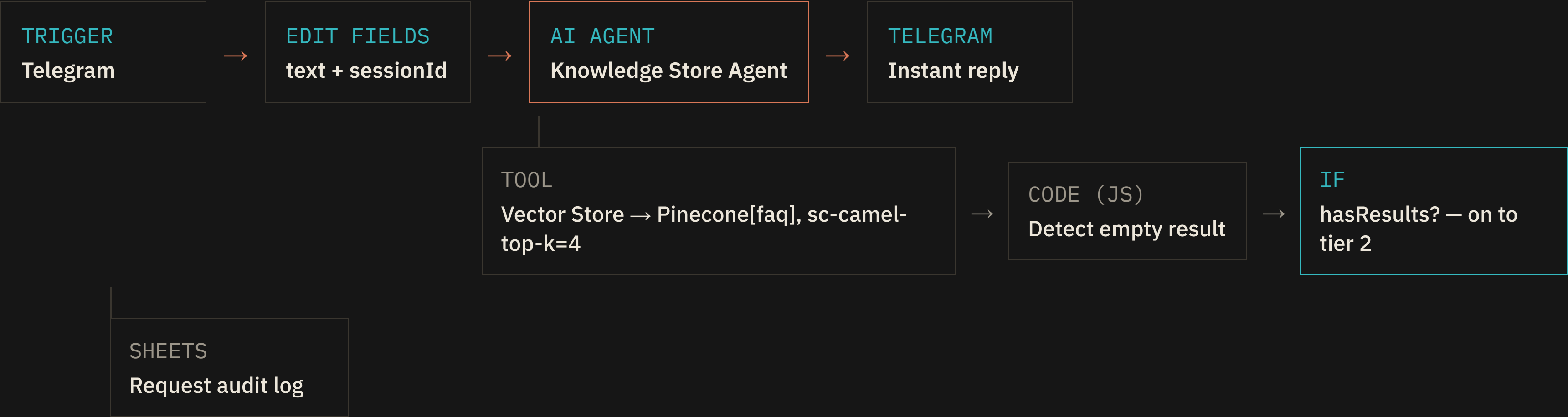
When the curated FAQ has no answer, there needs to be a path to the full document corpus.

Two retrieval tiers



Both namespaces — `faq` and `new2` — live in one Pinecone index, `legis`: vetted entries always outrank the raw corpus.

Fast answer from the curated FAQ

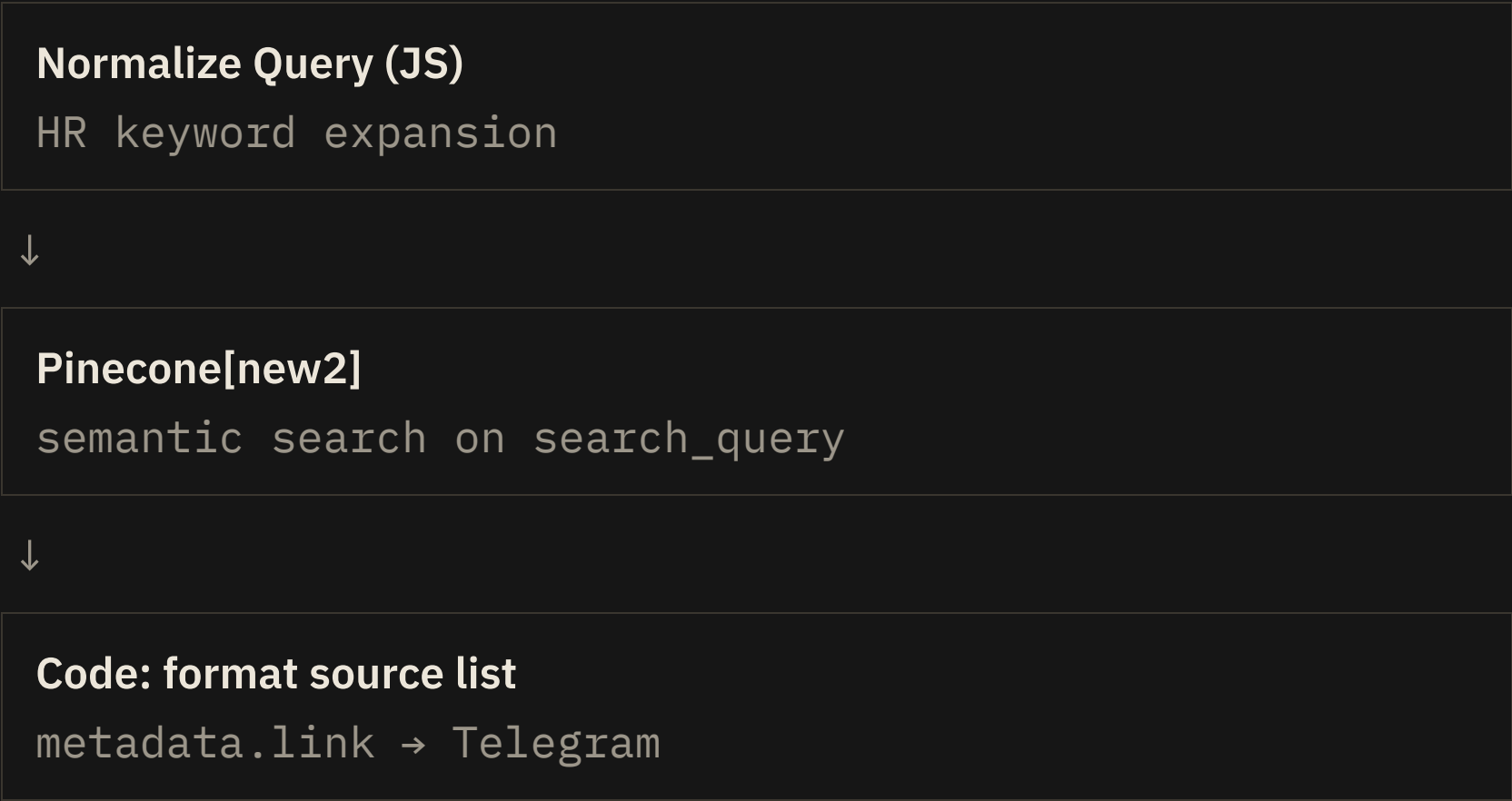


The reply and the empty-result check run in parallel — the user never waits on the second branch unless it's needed.

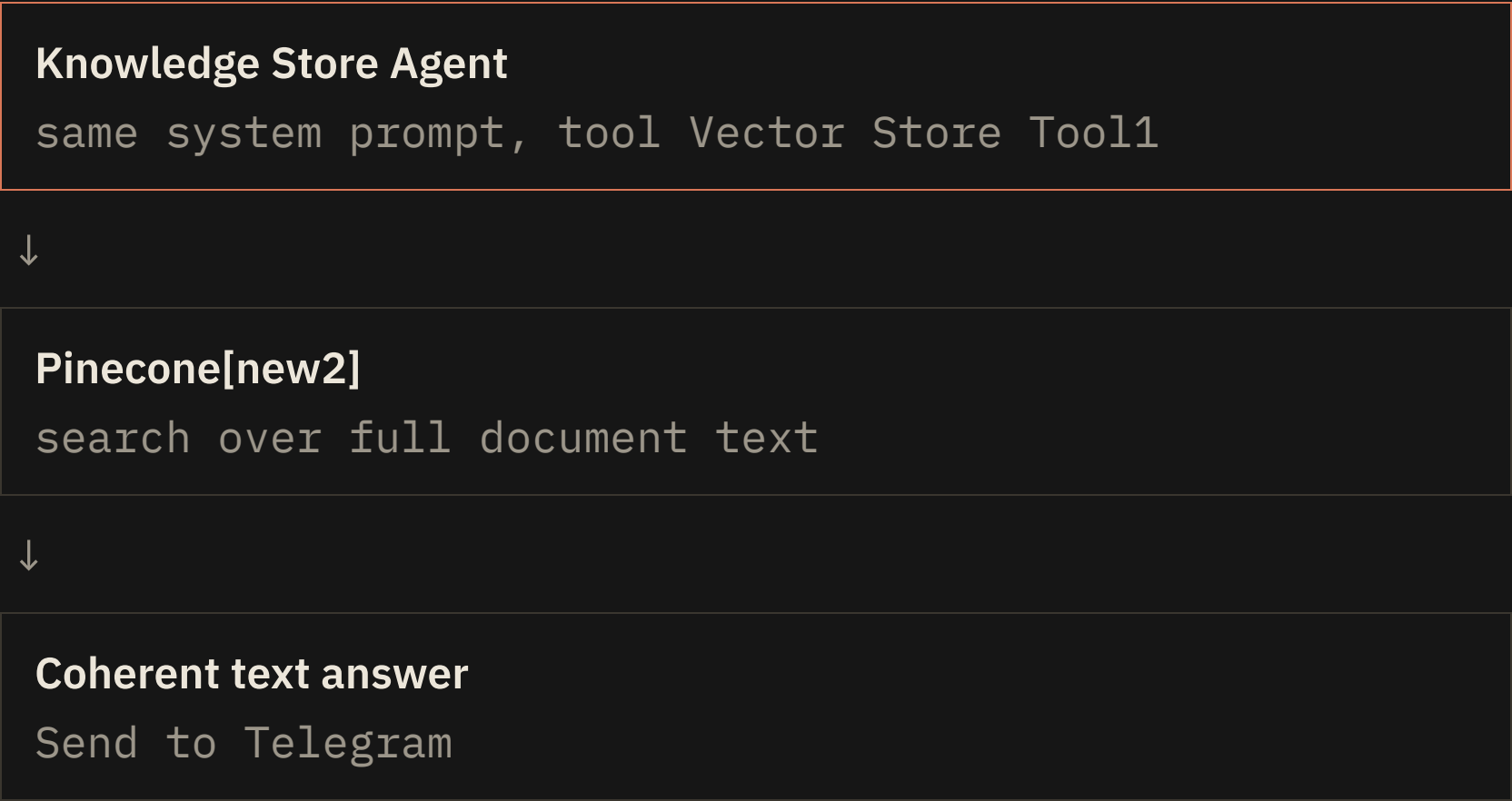
Escalation to the full document corpus



DIRECT VECTOR SEARCH

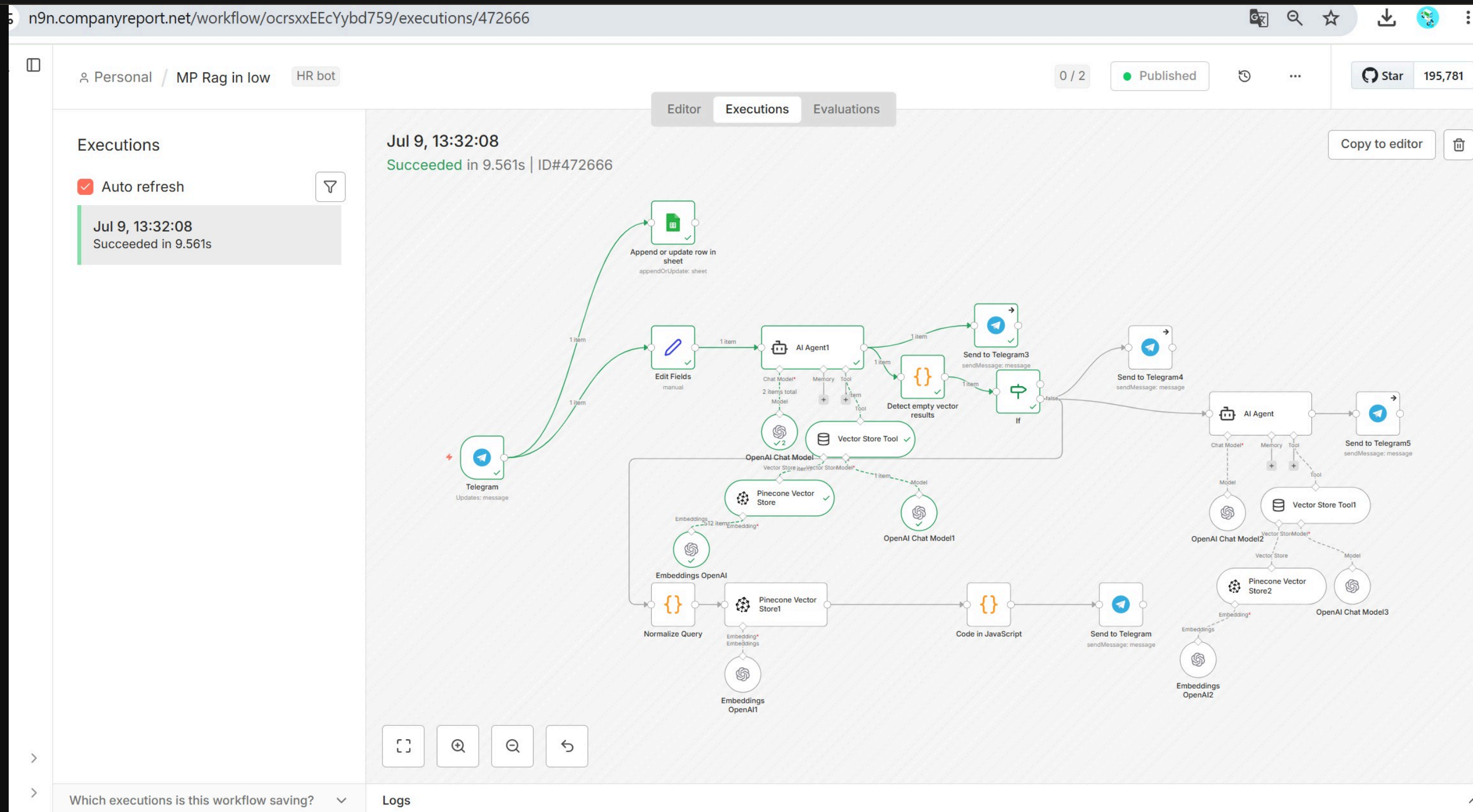


SECOND AI AGENT



06 - IMPLEMENTATION

The pipeline in n8n



Production workflow · Published · executions log

Key engineering decisions

01 Anti-hallucination guardrail

The prompt hard-codes the refusal phrase — a single, reliably parseable signal for the code downstream.

03 Asymmetric response timing

The FAQ answer ships immediately; the full-corpus follow-up arrives as a second message, only when needed.

05 Google Sheets over a database

A lightweight audit log replaces Postgres/Supabase — faster to stand up, sufficient for current volume.

02 Classifier outside the LLM

The empty-result check is a cheap JS node scanning for marker phrases, not a second model call.

04 Namespaces, not indexes

faq and new2 share one index — simpler to operate, while preserving source priority.

06 sessionId = chat_id

Per-user conversational memory for the agent, with zero extra infrastructure.

Tech stack

Orchestration	n8n · production workflow (active)
Interface	Telegram Bot API
LLM	OpenAI gpt-4.1-mini · 2 independent AI Agents
Vector search	Pinecone · index legis · namespace faq / new2
Embeddings	OpenAI Embeddings · 512 dim
Logging	Google Sheets · append/update per request

An instant FAQ answer — or sources and a full-corpus response

Users always get a result: fast and vetted, or — when no ready answer exists — a grounded response synthesized from the full body of regulatory documents.